

SAGE LAB · MECHANISTIC INTERPRETABILITY

Cross-Layer Transcoders for Protein Language Models

Scaling sparse circuit tracing to ESM-2 650M for zero-shot mutation-effect prediction.

Sagnik Chatterjee — Graduate Student Researcher, SAGE Lab

Advisors: Saishradha Mohanty · Anna Green · University of Massachusetts Amherst

Working session with Xiaocong Yang (UIUC) · 2 July 2026

About me

- MS in Computer Science, UMass Amherst (2025–2027); BE in Computer Science, Manipal Institute of Technology.
- Graduate Student Researcher, SAGE Lab (May 2026–present) — mechanistic interpretability for protein language models.
- Prior UMass research: CIIR Lab (typed recursive-agent frameworks) and BioNLP Lab (LoRA fine-tuning of LLaMA-3, +6.25pp on USMLE).
- Three 2026 preprints: small-LM reasoning, continual tool-use, and biological sequence representation (SCOPE operon embeddings).
- Three years as Backend & Integration engineer at IBM — distributed systems, Kafka pipelines, observability at scale.

What I am building in the SAGE Lab

- A cross-layer transcoder (CLT) for ESM-2 650M enabling unsupervised zero-shot mutation-effect prediction.
- Extending mechanistic interpretability from language models to protein language models at scale.
- Building biological circuits from sparse CLT features to test whether ESM-2 650M encodes structurally and evolutionarily meaningful representations.
- Codebase: a fork of EleutherAI's sparsify, extended with CLT support, tensor parallelism, bfloat16, and a KL+FVU loss.

Why mechanistic interpretability for proteins?

- ESM-2 predicts structure and function from sequence and underpins modern computational biology — but how it computes contacts and functional sites is opaque.
- Causal circuits yield scientific hypotheses, not just predictions — e.g. “a catalytic-motif feature at layer 6 gates a domain-recognition feature at layer 20.”
- Frontier: Nainani et al. 2025 (SAE latents causally mediate contact prediction) and ProtoMech 2026 (CLTs recover 82–89% at ESM2-8M/35M).
- The gap: nobody has trained CLTs at 650M — rich enough to matter biologically, but the training cost explodes. That is this project.

SAE → Transcoder → Cross-Layer Transcoder

	reads		predicts (reconstructs)
SAE	act at L	→	same act at L
Transcoder	input of L	→	OUTPUT of L
CLT (mine)	input of L	→	outputs of L, L+1, ..., L+N

- Real biological computation flows across layers. A CLT encoder fires once at layer L; its decoder emits into every downstream target — the learned dictionary is the cross-layer wiring.
- TopK activation (Gao 2024) fires exactly k latents per token, no L1 tuning. Expansion $128 \times d_model\ 1280 = 163,840$ latents per layer.

Progress so far — what runs today

- ESM-2 650M integrated into the sparsify pipeline: CLS-token handling, no-BOS, hookpoints before layernorm.
- Cross-layer transcoder training working — configurable reach and four coalescing strategies (none, concat, per-layer, group).
- KL + FVU training implemented — reconstruct well and stay faithful to ESM-2's computation.
- Tensor parallelism via DTensor plus bfloat16 for weights that exceed a single GPU.
- Sweep analysis ranking W&B runs by FVU, replacement-CE increase, L0, and dead-latent percentage.
- Current focus (Objective 1): make 650M CLT training fast and memory-tractable so the recipe can scale.

Why is 650M CLT training slow?

Bottleneck	What happens
1 · No activation caching	Full frozen ESM-2 650M forward pass every step (two passes for KL-FVU); every sweep re-runs ESM-2 from scratch.
2 · Enormous CLT size	At EF=128, cross_layer=12: ~89B params across 33 layers — larger than ESM-2 itself, ~121× the 35M recipe.
3 · Scale & comms	TopK over 163,840 latents × ~32K tokens; DTensor all-gather each layer; custom backward blocks torch.compile.

Efficiency roadmap (impact → effort)

- Profile first — torch.profiler / nsys to split time between ESM-2 forward and CLT before changing anything.
- Activation caching — precompute and memory-map ESM-2 activations; expected 3–10× speedup, sweeps become near-free.
- Shrink the CLT — start EF=32, cross_layer=4–6, layer_stride=2; reproduce ProtoMech-quality at reduced scale first.
- LoRA-style factorization — shared base dictionary + small per-target low-rank decoder adapters; low-rank encoder.
- Loop-level wins — torch.compile coverage, 8-bit Adam, activation checkpointing, TP comms overlap.

What “tractable at 650M” means, measured

Metric	Target	Meaning
val mean FVU	< 0.05	reconstruction quality
replacement CE increase	< 0.1 nats	faithfulness to ESM-2
dead latents	< 5%	dictionary utilization
memory / throughput	< 40 GB/GPU, 3–10×	cheap enough to sweep

Then: go after the biology

- Reproduce and sharpen Nainani's motif→domain contact circuit with CLTs (actual cross-layer wiring) via activation patching.
- Circuit discovery at 650M — ProtoMech-style greedy latent selection + integrated gradients. Do larger models use the same circuits at higher resolution?
- Causal validation — ProteinGym: test whether circuit-relevant CLT features alone predict mutation effects.
- Scale out — carry the optimized pipeline to ESM-C 3B–6B, the target that makes the efficiency work worth doing now.

Training works. Now make it scale.

- Done — CLT + KL-FVU + tensor-parallel training running on ESM-2 650M.
- Now — profile, cache activations, shrink and factorize the CLT to hit the target gates.
- Next — motif→domain circuits at 650M, ProteinGym validation, scale to ESM-C 3B–6B.